MECHANICAL ENGINEERING / SELECTED WORK

Tanmay Patil

From CAD to working equipment.

Mechanical engineer combining design, hands-on manufacturing, and troubleshooting to improve how equipment is built, assembled, and maintained.



Interlocking differential stand - fabricated hardware

Mechanical Design | **Manufacturing** | **Test & Validation**

75%
less service time
Die-box redesign: 16 h to 4 h

25%
lighter drivetrain
Chain drive vs. previous prop shaft

50%
less assembly time
Chain drive vs. previous system

FEATURED PROJECTS

01 Bar-shear die-box redesign
02 ATV chain-drive system
03 CVT / ECVT modeling & DAQ
04 PLC & ABB robot troubleshooting
05 Dust-collector winch damping
06 Gauging HMI migration

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PROJECT 01 / MID-WEST FORGE

Designing for a shorter service stop

Redesigned a die-box assembly to address recurring bolt shear failures and improve serviceability.



THE PROBLEM

Recurring bolt shear failures in a bar-shear die-box assembly created reliability and maintenance problems. Service work took 16 hours with the previous arrangement.

MY CONTRIBUTION

Redesigned the mechanical assembly using design-for-manufacture and design-for-assembly principles, with attention to service access and maintainability.

APPROACH

- Investigated recurring fastener failures and the assembly's serviceability constraints.
- Developed a revised mechanical assembly suited to manufacturing and maintenance.
- Focused the redesign on reliability and the work required for a service intervention.

RESULT

Service time fell from 16 hours to 4 hours: a 75% reduction.

TOOLS & METHODS SolidWorks / DFM / DFA / Mechanical redesign / Failure analysis

PROJECT 02 / BAJA SAE DRIVETRAIN DEVELOPMENT

A lighter drivetrain, faster to assemble

Developed a chain-drive system to replace the previous prop-shaft arrangement, reducing weight and assembly time.

CHAIN DRIVE COMPARED WITH PREVIOUS PROP-SHAFT SYSTEM

25%

lighter system

50%

less assembly time

System-level comparison against the earlier drivetrain arrangement.

THE PROBLEM

The previous prop-shaft system provided the baseline for a lighter, more serviceable drivetrain arrangement.

MY CONTRIBUTION

Worked on the chain-drive system's mechanical design and assembly improvements as part of Baja SAE drivetrain development.

APPROACH

- Evaluated a chain-drive arrangement against the earlier prop-shaft system.
- Used CAD and engineering analysis to develop the drivetrain layout.
- Applied assembly-focused design decisions to reduce the work required to assemble the system.

RESULT

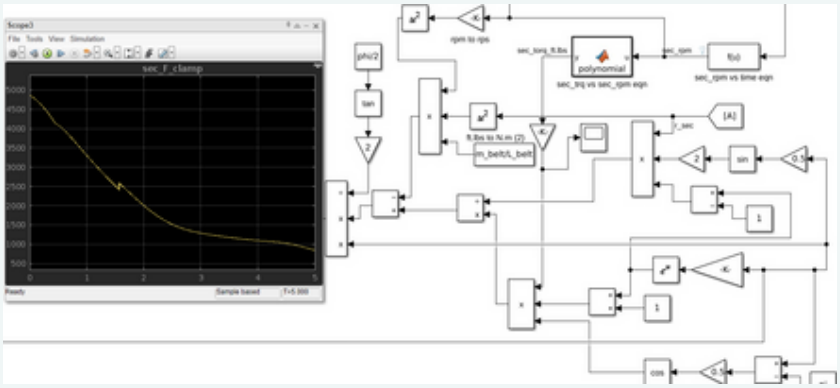
The chain-drive system was 25% lighter and took 50% less time to assemble than the previous prop-shaft system.

TOOLS & METHODS SolidWorks / Drivetrain design / DFM / DFA / Engineering analysis

PROJECT 03 / CVT / ECVT DEVELOPMENT

Bringing a CVT model to the test vehicle

Built a data-acquisition system to compare real drivetrain behavior with a CVT simulation.



Earlier Simulink CVT model from my design portfolio. The DAQ build described here is not pictured.

THE PROBLEM

A CVT model needed to be compared with physical drivetrain behavior, using measurements that describe both transmission motion and rotational speeds.

MY CONTRIBUTION

Built a DAQ system to collect sheave linear travel, engine RPM, and wheel-speed RPM, and compared these measurements with a CVT model I developed.

APPROACH

- Developed a CVT model in MATLAB / Simulink.
- Built instrumentation to capture sheave displacement alongside engine and wheel rotational speeds.
- Compared measured signals with the model to assess agreement and investigate differences.

RESULT

Produced a working measurement-to-model comparison across sheave travel, engine speed, and wheel speed.

TOOLS & METHODS MATLAB / Simulink / Data acquisition / Sensors / Model comparison

PROJECT 04 / MID-WEST FORGE

Tracing faults across mechanical and controls systems

Combined PLC logic tracing, electrical diagnosis, and robot maintenance to investigate production equipment faults.

EXPERIENCE ACROSS CONNECTED EQUIPMENT SYSTEMS

PLC logic	Wiring	Axis drives	Robot control
RSLinx / RSLogix 500	Pendant communication	Axis-specific short-circuit fault	S4C board / rev-counter work

Examples of hands-on support; each fault was investigated in its own equipment context.

THE PROBLEM

Production equipment issues can originate in wiring, controls, drives, or mechanical components. The symptom alone does not identify the source.

MY CONTRIBUTION

Used RSLinx and RSLogix 500 for PLC troubleshooting. On ABB 6400 robots with S4C controllers, performed rev-counter updates for individual axes and diagnosed pendant wiring, axis-drive, and controller-board problems.

APPROACH

- Traced control logic and electrical faults using RSLinx and RSLogix 500.
- Identified sliced pendant wires associated with communication issues.
- Investigated an axis-specific short-circuit error involving a failing drive.

RESULT

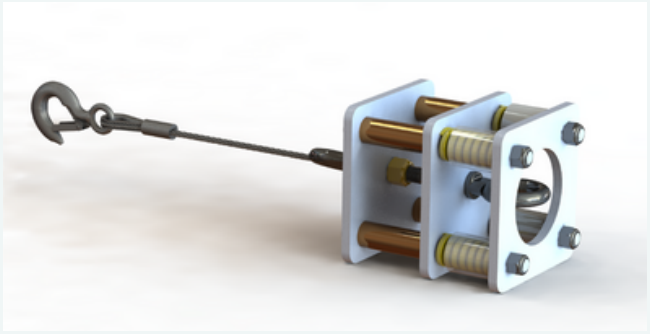
Identified wiring, axis-drive, and controller-board faults; performed individual-axis rev-counter updates.

TOOLS & METHODS RSLinx / RSLogix 500 / ABB 6400 / S4C controller / Electrical troubleshooting

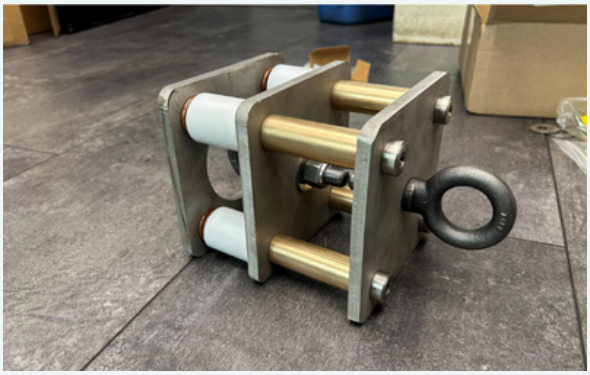
PROJECT 05 / MID-WEST FORGE

From a repeated failure to custom hardware

Developed a custom damping arrangement for a dust-collector hopper winch experiencing wire-rope failures.



Custom damping assembly - CAD rendering.



Fabricated custom damping assembly.

THE PROBLEM

The hopper winch arrangement needed to address recurring wire-rope snapping and improve the way the mechanism operated.

MY CONTRIBUTION

Investigated the problem, performed calculations, and designed a custom damping system. The portfolio documents both the CAD design and fabricated hardware.

APPROACH

- Investigated the recurring failure and developed a mechanical solution.
- Performed design calculations and created the custom damping assembly in CAD.
- Carried the concept through to fabricated hardware.

RESULT

Designed and fabricated a custom damping assembly for the hopper winch.

TOOLS & METHODS Mechanical calculations / SolidWorks / Prototyping / Fabrication

PROJECT 06 / MID-WEST FORGE

Moving a gauging screen beyond a legacy HMI

Recreating a gauging screen in EZTouch 2.5 to move away from a legacy HMI.

CURRENT PROJECT

In progress

Legacy gauging interface → EZTouch 2.5

Recreating the operator-facing gauging screen for an EZAutomation HMI.

THE PROBLEM

The gauging interface is being recreated as part of a move away from a legacy HMI.

MY CONTRIBUTION

Currently recreating the gauging screen using EZTouch 2.5 for an EZAutomation HMI.

APPROACH

- Review the existing gauging-screen functions.
- Recreate the operator-facing screen in EZTouch 2.5.

CURRENT STATUS

Work is ongoing. A completed deployment or production result is not claimed.

TOOLS & METHODS EZTouch 2.5 / EZAutomation HMI / Operator interface

CAPABILITIES / BACKGROUND / CONTACT

Design, build, diagnose.

My work connects mechanical design with the practical demands of production. At Mid-West Forge, I work on equipment redesign, rebuilds, and mechanical, electrical, and controls troubleshooting. Earlier drivetrain projects developed my experience in CAD, simulation, fabrication, instrumentation, and testing. I bring those perspectives together when diagnosing a fault or developing a design that needs to work on the shop floor.

MECHANICAL DESIGN

SolidWorks & PDM / Siemens NX / GD&T / DFM / DFA / Mechanism design / Engineering drawings / Prototyping & fabrication

MANUFACTURING & CONTROLS

Root-cause analysis / Equipment rebuilds / RSLinx / RSLogix 500 / EZTouch 2.5 / ABB 6400 / S4C / Electrical & pneumatic troubleshooting / PFMEA

ANALYSIS & TESTING

MATLAB / Simulink / ANSYS FEA / Altair HyperWorks / Inspire / ADAMS / MotionView / Data acquisition / Sensor integration / Model-to-test comparison

EXPERIENCE

Design and Rebuild Engineer | Mid-West Forge

May 2025 - Present

Engineering Intern | Mid-West Forge

May - August 2024

EDUCATION

Master of Science in Mechanical Engineering

University of Michigan, Ann Arbor | May 2025

Bachelor of Science in Mechanical Engineering

K.J. Somaiya College of Engineering, Mumbai | May 2023

CONNECT

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